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Brake apparatus for vehicle

Abstract

A brake apparatus for a vehicle may include: a caliper body module provided with a finger part and a cylinder part spaced apart from each other and facing each other, a pad plate part disposed between the finger part and the cylinder part, and configured to selectively come into contact with a disk module rotated together with a rotor to generate a braking force, a piston module installed in the cylinder part to be movable forward and backward and configured to press the pad plate part, a guide member disposed on both sides of the caliper body module and configured to movably support the caliper body module, and a first fastening member extending from the guide member and fastened to a first mounting member provided in a stator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from and the benefit of Korean Patent Application No. 10-2021-0043272, filed on Apr. 2, 2021 and Korean Patent Application No. 10-2021-0047230, filed on Apr. 12, 2021, which are hereby incorporated by reference for all purposes as if set forth herein.

BACKGROUND

Field

(2) Exemplary embodiments of the present disclosure relate to a brake apparatus for a vehicle, and more particularly, to a brake apparatus for a vehicle provided with an in-wheel motor.

Discussion of the Background

(3) In general, a brake apparatus for a vehicle is an apparatus that pushes a piston through hydraulic pressure generated from a master cylinder or a rotational force of a motor to bring a pad and a disk into close contact with each other, and brakes a vehicle by using friction between the pad and the disk. As such a brake apparatus for a vehicle, a floating type caliper brake composed of a caliper body and a torque member is widely used.

(4) However, in an e-corner module vehicle that is a next-generation platform, an in-wheel driving device, which is mounted inside each wheel of the vehicle so that each wheel may be controlled by independent driving, is installed. Such an in-wheel itself is constituted by one driving motor and thus has a very narrow space for mounting a brake apparatus. Therefore, there is a problem in that it is difficult to satisfy internal layout conditions of the in-wheel and simultaneously secure braking safety equivalent to that of a general brake system through an existing mass-produced caliper brake.

(5) The background art of the present disclosure is disclosed in Korean Patent Application Laid-Open No. 10-2010-0098846 (published on Sep. 10, 2010 and entitled “Disk Brake having Parking Function”).

SUMMARY

(6) Various embodiments are directed to a brake apparatus for a vehicle, capable of satisfying layout conditions of an in-wheel motor and improving assembling efficiency of a caliper body.

(7) In order to solve the above problems, a brake apparatus for a vehicle in accordance with the present disclosure may include: a caliper body module provided with a finger part and a cylinder part spaced apart from each other and facing each other; a pad plate part disposed between the finger part and the cylinder part, and configured to selectively come into contact with a disk module rotated together with a rotor to generate a braking force; a piston module installed in the cylinder part to be movable forward and backward and configured to press the pad plate part; a guide member disposed on both sides of the caliper body module and configured to movably support the caliper body module; and a first fastening member extending from the guide member and fastened to a first mounting member provided in a stator.

(8) Furthermore, the first fastening member may be disposed on substantially the same axial line as the guide member.

(9) Furthermore, the first fastening member may protrude to an outside of the finger part by a predetermined distance along a longitudinal direction of the guide member.

(10) Furthermore, a first fastening reinforcing part may be provided on an outer periphery of the first fastening member to reinforce a fastening force of the first fastening member with respect to the first mounting member.

(11) Furthermore, a central axis of the guide member may be disposed at substantially the same

height as a gravity axis of the caliper body module.

(12) Furthermore, the guide member may include: a guide pin part configured to penetrate the caliper body module and disposed in a direction parallel to a forward and backward movement direction of the piston module; a friction reduction part configured to surround the guide pin part; a movement part slidably connected to the friction reduction part and configured to move the caliper body module by a pressing force of the piston module; and a blocking part configured to block foreign substances from entering between the caliper body module and the friction reduction part.

(13) Furthermore, the guide pin part may have one side connected through the finger part.

(14) Furthermore, the guide pin part may movably support the pad plate part.

(15) A brake apparatus for a vehicle in accordance with the present disclosure may include: a caliper body module provided with a finger part and a cylinder part spaced apart from each other and facing each other; a pad plate part disposed between the finger part and the cylinder part, and configured to selectively come into contact with a disk module rotated together with a rotor to generate a braking force; a piston module installed in the cylinder part to be movable forward and backward and configured to press the pad plate part; a guide member disposed on both sides of the caliper body module and configured to movably support the caliper body module; and a second fastening member extending from the guide member and fastened to a second mounting member provided in a knuckle.

(16) Furthermore, the second fastening member may be disposed on substantially the same axial line as the guide member.

(17) Furthermore, the second fastening member may protrude to an outside of the caliper body module by a predetermined distance along a longitudinal direction of the guide member.

(18) Furthermore, a second fastening reinforcing part may be provided on an outer periphery of the second fastening member to reinforce a fastening force of the second fastening member with respect to the second mounting member.

(19) Furthermore, a central axis of the guide member may be disposed at substantially the same height as a gravity axis of the caliper body module.

(20) Furthermore, the guide member may include: a guide pin part configured to penetrate the caliper body module and disposed in a direction parallel to a forward and backward movement direction of the piston module; a friction reduction part configured to surround the guide pin part; a movement part slidably connected to the friction reduction part and configured to move the caliper body module by a pressing force of the piston module; and a blocking part configured to block foreign substances from entering between the caliper body module and the friction reduction part.

(21) Furthermore, the guide pin part may have one side connected through the finger part.

(22) A brake apparatus for a vehicle in accordance with the present disclosure may include: a stator; a rotor disposed inside the stator and rotated when power is applied to the stator; a wheel rotated in connection with the rotor; a disk module interlocked with the rotor; an internal braking device mounted on the stator and configured to provide a frictional force to one side of the disk module; and an external braking device mounted on the stator and configured to provide a frictional force to the other side of the disk module.

(23) Furthermore, the stator may include: a housing unit mounted with the external braking device and fixedly coupled to a fixed unit; a stator body mounted on the housing unit and configured to surround the rotor and rotate the rotor when power is applied; and an internal space in the housing unit configured to accommodate the internal braking device.

(24) Furthermore, the internal braking device may include: an internal piston part mounted in the internal space and having a variable length; and an internal pad part mounted on the internal piston part and being in surface contact with the disk module according to a change in the length of the internal piston part.

(25) Furthermore, the internal braking device may include: an internal case mounted in the internal space and coupled to the external braking device; an internal variable part mounted in the internal

case and having a variable length; and an internal contact part mounted on the internal variable part and being in surface contact with the disk module according to a change in the length of the internal variable part.

(26) Furthermore, the external braking device may include: an external case coupled to the stator; an external variable part mounted in the external case and having a variable length; and an external contact part mounted on the external variable part and being in surface contact with the disk module according to a change in the length of the external variable part.

(27) According to a brake apparatus for a vehicle in accordance with the present disclosure, a first fastening member is fastened to a first mounting member provided in a stator to fix a caliper body module, thereby substantially preventing the caliper body module from being kept in an unassembled state in a process in which a rotator, the stator, and a disk module are coupled to a vehicle body or a knuckle.

(28) Furthermore, according to a brake apparatus for a vehicle in accordance with the present disclosure, the first fastening member and a second fastening member may be disposed on substantially the same axial line as a guide member and may play a role of a torque member formed in an existing brake caliper on substantially the same axial line as the guide member, thereby substantially preventing interference with parts inside an in-wheel.

(29) Furthermore, according to a brake apparatus for a vehicle in accordance with the present disclosure, a guide pin part may be connected through a finger part, thereby sufficiently securing the rigidity of the finger part and reducing the thicknesses of the caliper body module in a height direction and a longitudinal direction at the same time.

(30) Furthermore, according to a brake apparatus for a vehicle in accordance with the present disclosure, a central axis of the guide member may be disposed at substantially the same height as a gravity axis of the caliper body module, thereby reducing a moment generated in the caliper body module when the caliper body module slides and thus substantially preventing vibration and noise from being generated.

(31) Furthermore, according to a brake apparatus for a vehicle in accordance with the present disclosure, an internal braking device and an external braking device may be fixed to the stator, thereby improving the assembling efficiency of the vehicle body and securing the rigidity of the internal braking device and the external braking device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-sectional view schematically illustrating an installation state of a brake apparatus for a vehicle in accordance with a first embodiment of the present disclosure.
- (2) FIG. 2 is a perspective view schematically illustrating the configuration of the brake apparatus for a vehicle in accordance with the first embodiment of the present disclosure.
- (3) FIG. 3A and FIG. 3B are a side view and a front view schematically illustrating the configuration of the brake apparatus for a vehicle in accordance with the first embodiment of the present disclosure.
- (4) FIG. 4 is a cross-sectional view schematically illustrating the configuration of a guide member in accordance with the first embodiment of the present disclosure.
- (5) FIG. 5 is a cross-sectional view schematically illustrating a state in which the guide member in accordance with the first embodiment of the present disclosure supports a pad plate part.
- (6) FIG. 6 is an enlarged view schematically illustrating the configuration of a first fastening member in accordance with the first embodiment of the present disclosure.
- (7) FIG. 7 is a view schematically illustrating an installation state of a brake apparatus for a vehicle in accordance with a second embodiment of the present disclosure.

- (8) FIG. 8 is a perspective view schematically illustrating the configuration of the brake apparatus for a vehicle in accordance with the second embodiment of the present disclosure.
- (9) FIG. 9 is a cross-sectional view schematically illustrating the configuration of the brake apparatus for a vehicle in accordance with the second embodiment of the present disclosure.
- (10) FIG. 10 is a cross-sectional view illustrating a case where a second mounting part is disposed at a position different from that of FIG. 9.
- (11) FIG. 11 is a view schematically illustrating a brake apparatus for a vehicle in accordance with a third embodiment of the present disclosure.
- (12) FIG. 12 is a view schematically illustrating a stator in accordance with the third embodiment of the present disclosure.
- (13) FIG. 13 is a view schematically illustrating a first modification of an internal braking device in accordance with the third embodiment of the present disclosure.
- (14) FIG. 14 is a view schematically illustrating a second modification of the internal braking device in accordance with the third embodiment of the present disclosure.
- (15) FIG. 15 is a view schematically illustrating an external braking device in accordance with the third embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS

- (16) Hereinafter, embodiments of a brake apparatus for a vehicle in accordance with the present disclosure will be described with reference to the accompanying drawings.
- (17) In this process, the thicknesses of lines or the sizes of elements illustrated in the drawings may be exaggerated for the purpose of clarity and convenience of explanation. Furthermore, terms to be described later are terms defined in consideration of functions thereof in the present disclosure and may be changed according to the intention of a user or an operator, or practice. Accordingly, such terms should be defined based on the disclosure over the present specification.
- (18) Furthermore, in the present specification, when a certain part is referred to as being “connected (or coupled) to” another part, it may indicate that the former part is directly connected (or coupled) to the latter part or indirectly connected (or coupled) to the latter part with another part interposed therebetween. In the present specification, when a certain part “includes (or comprises)” a certain component, it means that the element does not exclude another component but may further “include (or comprise)” another component, unless referred to the contrary.
- (19) Furthermore, substantially the same reference numerals may refer to substantially the same components throughout the present specification. Even though substantially the same reference numerals or similar reference numerals are not mentioned or described in a specific drawing, the reference numerals may be described based on other drawings. Furthermore, even though there is a portion which is not indicated by reference numerals in a specific drawing, the portion may be described based on other drawings. Furthermore, the number, shapes, and sizes of detailed components included in the drawings of the present application and relative differences in the sizes are set for convenience of understanding, and do not limit embodiments and may be implemented in various forms.
- (20) FIG. 1 is a cross-sectional view schematically illustrating an installation state of a brake apparatus 1 for a vehicle in accordance with a first embodiment of the present disclosure.
- (21) Referring to FIG. 1, a rotor 20 and a stator 30 are installed inside a wheel 10 of a vehicle and generate a driving force and rotate the wheel 10.
- (22) The rotor 20 is installed inside the wheel 10 and is rotated under the influence of a magnetic field formed by the stator 30 to be described below. The rotor 20 in accordance the present embodiment is formed in a cylindrical shape having a “C”-shaped cross-section with one side open, and is disposed so that its outer peripheral surface faces an inner peripheral surface of the wheel 10. The rotor 20 is disposed so that its central axis is located on substantially the same axial line at the central axis of the wheel 10, and is connected to the wheel 10 via a wheel bearing 5. The rotor 20 is configured to include a magnetic body such as a permanent magnet or metal, is rotated under the

influence of a magnetic field formed by the stator **30**, and rotates the wheel **10**.

(23) The stator **30** is installed inside the rotor **20**, and receives power from an exterior to form a magnetic field that rotates the rotor **20**. The stator **30** in accordance with the present embodiment is formed in the form of a hollow ring having an “L”-shaped cross-section, and is disposed so that its outer peripheral surface faces an inner peripheral surface of the rotor **20**. The stator **30** is configured to include an electromagnet and rotates the rotor **20** by an electromagnet force generated as power is supplied from an exterior. A connector **38** connects the stator **30** to an external power source to supply power to the stator **30**. The stator **30** is fixed to a knuckle **6**, a motor bracket (not illustrated), or the like fixed to a vehicle body, and is restricted from rotating inside the wheel **10**.

(24) The stator **30** is provided with a first mounting member **30a** that fixes the brake apparatus **1** for a vehicle in accordance with the present embodiment to the stator **30**. A detailed shape of the first mounting member **30a** will be described below.

(25) A disk module **40** is provided to be rotatable together with the rotor **20**. The disk module **40** in accordance with the present embodiment is formed to have a disk shape and is disposed inside the stator **30**. The disk module **40** is connected to the wheel **10** and the rotor **20** via the wheel bearing **5** and is rotated together with the rotor **20** when the rotor **20** is rotated. A diameter of the disk module **40** can be variously changed in design according to a diameter and the like of the wheel **10**.

(26) The brake apparatus **1** for a vehicle in accordance with the present embodiment is disposed between the stator **30** and the disk module **40**, is provided to be able to allow or limit the rotation of the disk module **40**, and generates a braking force in the vehicle. The brake apparatus **1** for a vehicle is fixed to the stator **30** via the first mounting member **30a** described above.

(27) FIG. **2** is a perspective view schematically illustrating the configuration of the brake apparatus **1** for a vehicle in accordance with the first embodiment of the present disclosure, and FIG. **3A** and FIG. **3B** are a side view and a front view schematically illustrating the configuration of the brake apparatus **1** for a vehicle in accordance with the first embodiment of the present disclosure.

(28) Referring to FIG. **1** to FIG. **3**, the brake apparatus **1** for a vehicle in accordance with the present embodiment includes a caliper body module **100**, a pair of pad plate parts **200**, a piston module **300**, a guide member **400**, and a first fastening member **500**.

(29) The caliper body module **100** forms a schematic appearance of the brake apparatus **1** for a vehicle. The caliper body module **100** in accordance with the present embodiment includes a bridge part **110**, a finger part **120**, a cylinder part **130**, and an accommodating part **140**.

(30) The bridge part **110** forms an upper appearance of the caliper body module **100**, and supports the finger part **120**, the cylinder part **130**, and the accommodating part **140** as a whole, which will be described below.

(31) The finger part **120** vertically extends downward from a front end of the bridge part **110**. The finger part **120** is disposed so that its internal side surface faces one of the pair of pad plate parts **200**, which is disposed outside (left side in FIG. **1**) the disk module **40**. The finger part **120** slides by a pressing force of the piston module **300** to be described below, and presses the pad plate part **200** to the disk module **40** or releases the pressure. The finger part **120** may be integrally formed with the bridge part **110** by welding, press working, bending, or the like, and alternatively, may be detachably coupled to the bridge part **110** by bolting or the like.

(32) The cylinder part **130** vertically extends downward from the rear of the bridge part **110**. The cylinder part **130** is formed so that one side thereof is opened so as to provide a space where the piston module **300** to be described below is installed therein. The cylinder part **130** is disposed so that the opened side faces one of the pair of pad plate parts **200**, which is disposed inside (see FIG. **1**) the disk module **40**. The cylinder part **130** may be integrally formed with the bridge part **110** by welding, press working, bending, or the like, and alternatively, may be detachably coupled to the bridge part **110** by bolting or the like.

(33) The accommodating part **140** is provided as a pair and the pair of accommodating parts **140**

are disposed on both sides of the caliper body module **100**, respectively. A through hole is formed by horizontally penetrating the front and rear surfaces of the accommodating part **140** to provide a space into which the guide member **400** to be described below is inserted. The accommodating part **140** is slidably connected to the guide member **400**. The accommodating part **140** slides along the guide member **400** in a direction opposite to the forward and backward movement of the piston module **300**, and moves the caliper body module **100**. The accommodating part **140** may be integrally formed with the bridge part **110** by welding, press working, bending, or the like, and alternatively, may be detachably coupled to the bridge part **110** by bolting or the like.

(34) The pad plate part **200** is disposed between the finger part **120** and the cylinder part **130**, and selectively comes into contact with the disk module **40** rotated together with the rotor **20** to generate a braking force. The pad plate part **200** in accordance with the present embodiment is provided as a pair, and the pair of pad plate parts **200** are disposed to face the outer and inner side surfaces of the disk module **40**, respectively. The pair of pad plate parts **200** are installed with the disk module **40** interposed therebetween so as to be movable in a direction away from or close to the disk module **40**. In such a case, the pad plate parts **200** may be movably supported by the caliper body module **100**, the guide member **400**, and the like. A friction pad made of a material having a high frictional coefficient such as rubber may be attached to one surface of the pad plate part **200** facing the disk module **40**. A detailed shape of the pad plate part **200** can be variously changed in design within the technical spirit of a brake pad that comes into contact with the disk module **40** and applies a braking force to the vehicle.

(35) The piston module **300** is installed in the cylinder part **130** to move forward and backward and presses the pad plate parts **200**. The piston module **300** is disposed so that its one side faces one of the pair of pad plate parts **200**, which is disposed inside (see FIG. **1**) the disk module **40**. The piston module **300** is connected to an electric motor, receives a rotational force from the electric motor, and moves forward and backward with respect to the pad plate parts **200**. The piston module **300** protrudes in front of the cylinder part **130** when moving forward and presses the pad plate parts **200**. The piston module **300** slides the caliper body module **100** in a direction opposite to the forward/backward movement direction of the piston module **300** along the longitudinal direction of the guide member **400** by a reactional force generated by pressing the pad plate parts **200**. The piston module **300** returns to the inside of the cylinder part **130** when moving backward and releases the pressure on the pad plate parts **200**. The piston module **300** in accordance with the present embodiment may include a spindle that is rotated by the electric motor and has threads provided on the outer circumferential surface thereof, a nut that is screwed to the spindle to move linearly along the longitudinal direction of the spindle when the spindle is rotated, and a piston that moves forward and backward with respect to the pad plate parts **200** by a pressing force applied during the linear motion of the nut and pressures the pad plate parts **200** or releases the pressure on the pad plate parts **200**.

(36) The guide member **400** is disposed on both sides of the caliper body module **100** and movably supports the caliper body module **100**. That is, the guide member **400** is installed inside the accommodating part **140** and is provided to guide the slide movement of the caliper body module **100** due to the reactional force by the piston module **300**. The guide member **400** is provided as a pair, and the pair of guide members **400** are installed inside the pair of accommodating parts **140**, respectively. A central axis C of the guide member **400** may be disposed at substantially the same height as a gravity axis M of the caliper body module **100**. Accordingly, the guide member **400** can reduce a moment generated in the caliper body module **100** when the caliper body module **100** slides, thereby substantially preventing vibration and noise from being generated.

(37) FIG. **4** is a cross-sectional view schematically illustrating the configuration of the guide member **400** in accordance with the first embodiment of the present disclosure.

(38) Referring to FIG. **4**, the guide member **400** in accordance with the present embodiment includes a guide pin part **410**, a friction reduction part **420**, a movement part **430**, and a blocking

part **440**.

(39) The guide pin part **410** penetrates the caliper body module **100** and is disposed in a direction parallel to the forward/backward movement direction of the piston module **300**. The guide pin part **410** in accordance with the present embodiment is formed in a rod shape having a circular cross-section. A front end of the guide pin part **410** is inserted into a rear surface of the accommodating part **140**, so that the guide pin part **410** horizontally penetrates the front and rear surfaces of the accommodating part **140**. The central axis C of the guide pin part **410** is disposed at substantially the same height as the gravity axis M of the caliper body module **100**. A diameter of the guide pin part **410** may have a value smaller than that of the through hole of the accommodating part **140**. The guide pin part **410** may be provided at the rear end thereof with a head part that supports the guide pin part **410** in contact with the rear surface of the accommodating part **140** when the guide pin part **410** is completely inserted into the accommodating part **140**. The front end of the guide pin part **410** may be connected through the finger part **120** to reinforce the support rigidity of the finger part **120** extending from the bridge part **110**. Accordingly, the guide pin part **410** may reduce the thicknesses of the bridge part **110** in the height direction and the longitudinal direction, so that the size of the caliper body module **100** may be designed to fit the layout of an in-wheel.

(40) The friction reduction part **420** is disposed to surround the guide pin part **410** and supports the movement part **430**, which will be described below, to be slidably movable. The friction reduction part **420** in accordance with the present embodiment is formed in a hollow cylindrical shape and is closely fixed to an outer circumferential surface of the guide pin part **410**. The friction reduction part **420** may have a smooth surface or include a material having a low frictional coefficient. The friction reduction part **420** may be manufactured separately from the guide pin part **410** and assembled to the guide pin part **410**, and alternatively, may be installed by being integrally applied onto the outer circumferential surface of the guide pin part **410**.

(41) One side of the movement part **430** is slidably connected to the friction reduction part **420** and the other side thereof is connected to the caliper body module **100**. The movement part **430** slides along the longitudinal direction of the guide pin part **410** by the pressing force of the piston module **300**, and moves the caliper body module **100**. The movement part **430** in accordance with the present embodiment is formed in a hollow cylindrical shape and is disposed to surround the friction reduction part **420**. An inner circumferential surface of the movement part **430** is slidably in contact with an inner circumferential surface of the friction reduction part **420**. An outer inner circumferential surface of the movement part **430** is fixed to an inner circumferential surface of the accommodating part **140**.

(42) The blocking part **440** blocks foreign substances from entering between the caliper body module **100** and the friction reduction part **420**. The blocking part **440** in accordance with the present embodiment is formed in a hollow cylindrical shape and is disposed to surround an outer circumferential surface of the friction reduction part **420**. The blocking part **440** has one end fixed to the outer circumferential surface of the friction reduction part **420** and the other end fixed to the inner circumferential surface of the accommodating part **140**, thereby sealing between the accommodating part **140** and the friction reduction part **420**. When the caliper body module **100** slides, the other end of the blocking part **440** moves together with the caliper body module **100** in a state in which the one end thereof is fixed to the friction reduction part **420**. In such a case, the blocking part **440** may be formed to have a wavy structure or a bellows structure in cross section so as to be stretchable in the longitudinal direction thereof. Accordingly, the blocking part **440** may not interfere with the slide movement of the caliper body module **100** and substantially maintain foreign substance blocking performance at the same time. The blocking part **440** may be provided as a pair and the pair of blocking parts **440** may be disposed on both front and rear sides of the accommodating part **140**, respectively.

(43) The guide member **400** in accordance with the present embodiment may be provided to movably support the pad plate parts **200**.

(44) FIG. 5 is a cross-sectional view schematically illustrating a state in which the guide member **400** in accordance with the first embodiment of the present disclosure supports the pad plate part **200**.

(45) Referring to FIG. 5, the accommodating part **140** and the friction reduction part **420** are each formed to have a length so that the front ends thereof may be located behind the pad plate part **200** disposed behind (right side in FIG. 5) the disk module **40**.

(46) The guide pin part **410** horizontally extends in front of the accommodating part **140**, and the front end of the guide pin part **410** is connected through the finger part **120**. Accordingly, the outer circumferential surface of a front portion of the guide pin part **410** is exposed to an exterior.

(47) The pair of pad plate parts **200** are slidably mounted on the outer circumferential surface of the front portion of the guide pin part **410** exposed to an exterior, and are moved in the front-rear direction along the longitudinal direction of the guide pin part **410**.

(48) The first fastening member **500** extends from the guide member **400**. The first fastening member **500** is detachably fastened to the first mounting member **30a** provided in the stator **30**, and supports the caliper body module **100**. Accordingly, the first fastening member **500** may substantially prevent the caliper body module **100** from being kept in an unassembled state inside the wheel **10** until the rotator **20**, the stator **30**, and the disk module **40** are fixed to the vehicle body or the knuckle **6**. The first fastening member **500** may be disposed on substantially the same axial line as the guide member **400**. Accordingly, the first fastening member **500** may play a role of a torque member, which is formed in an existing brake caliper, on substantially the same axial line as the guide member **400**, thereby more easily satisfying the internal layout conditions of the in-wheel.

(49) The first mounting member **30a** vertically protrudes from the inner side surface of the stator **30** disposed in parallel to the radial direction of the wheel **10**. The first mounting member **30a** may be formed in a hollow shape with one side open so that the first fastening member **500** may be inserted therein. The opened side of the first mounting member **30a** may be disposed to face the first fastening member **500** and the central axis thereof may be disposed on substantially the same axial line as the central axis C of the guide member **400**.

(50) FIG. 6 is an enlarged view schematically illustrating the configuration of the first fastening member **500** in accordance with the first embodiment of the present disclosure.

(51) Referring to FIG. 1, FIG. 2, and FIG. 6, the first fastening member **500** in accordance with the present embodiment may be formed in a rod shape extending horizontally from the front end of the guide pin part **410** along the longitudinal direction of the guide member **400**. The central axis of the first fastening member **500** is disposed on substantially the same axial line as the central axis C of the guide pin part **410**. The first fastening member **500** protrudes to the outside of the finger part **120** by a predetermined distance as the front end of the guide pin part **410** penetrates the finger part **120**. The protruding distance of the first fastening member **500** may be changed in design to various values according to a distance between the front end of the guide pin part **410** and the first mounting member **30a**, and the like. The first fastening member **500** is inserted into the first mounting member **30a** while the rotator **20**, the stator **30**, and the disk module **40** are assembled into the wheel **10**, and is fastened to the first mounting member **30a** to fix the position of the caliper body module **100**.

(52) A first fastening reinforcing part **501** may be provided on an outer periphery of the first fastening member **500** to reinforce a fastening force of the first fastening member **500** with respect to the first mounting member **30a**. The first fastening reinforcing part **501** in accordance with the present embodiment may be formed in the form of threads that protrude from the outer circumferential surface of the first fastening member **500** and extend along the longitudinal direction of the first fastening member **500**. In such a case, threads corresponding to the first fastening reinforcing part **501** may be formed on the inner circumferential surface of the first mounting member **30a**. The shape of the first fastening reinforcing part **501** is not limited to the

shape of the thread, and may be variously changed in design within the technical spirit of a shape capable of reinforcing the fastening force of the first fastening member **500** with respect to the first mounting member **30a**.

(53) Hereinafter, a configuration of a brake apparatus **2** for a vehicle in accordance with a second embodiment of the present disclosure will be described.

(54) In this process, for convenience of description, description overlapping the brake apparatus **1** for a vehicle in accordance with the first embodiment of the present disclosure will be omitted.

(55) FIG. **7** is a view schematically illustrating an installation state of the brake apparatus **2** for a vehicle in accordance with the second embodiment of the present disclosure, FIG. **8** is a perspective view schematically illustrating the configuration of the brake apparatus **2** for a vehicle in accordance with the second embodiment of the present disclosure, and FIG. **9** is a cross-sectional view schematically illustrating the configuration of the brake apparatus **2** for a vehicle in accordance with the second embodiment of the present disclosure.

(56) Referring to FIG. **7** to FIG. **9**, the brake apparatus **2** for a vehicle in accordance with the second embodiment of the present disclosure includes the caliper body module **100**, the pad plate part **200**, the piston module **300**, the guide member **400**, and a second fastening member **600**.

(57) A guide pin part **410** in accordance with the second embodiment of the present disclosure is disposed in a direction opposite to the guide pin part **410** in accordance with the first embodiment of the present disclosure.

(58) More specifically, in the guide pin part **410** in accordance with the second embodiment of the present disclosure, as a front end thereof penetrates the finger part **120** and then is inserted into a front surface of the accommodating part **140**, the front and rear ends thereof are disposed on the rear and front sides of the caliper body module **100**, respectively. Accordingly, the guide pin part **410** may cause the second fastening member **600** to be described below to extend toward a second mounting member **6a** formed on the rear side of the caliper body module **100**. A head part formed at the rear end of the guide pin part **410** is supported in contact with the outer side surface of the finger part **120**.

(59) As illustrated in FIG. **9**, as the length of the accommodating part **140** is formed to be shorter than that of the guide pin part **410**, the guide pin part **410** may movably support the pad plate part **200**. More specifically, as the pad plate part **200** is slidably mounted on the outer circumferential surface of a rear portion of the guide pin part **410** exposed to an exterior, the guide pin part **410** may guide the movement of the pad plate part **200**.

(60) The second fastening member **600** extends from the guide member **400**. The second fastening member **600** is detachably fastened to the second mounting member **6a** provided in the knuckle **6**, and supports the caliper body module **100**. The second fastening member **600** may be disposed on substantially the same axial line as the guide member **400**. Accordingly, the second fastening member **600** may play a role of a torque member, which is formed in an existing brake caliper, on substantially the same axial line as the guide member **400**, thereby more easily satisfying the internal layout conditions of the in-wheel.

(61) The second mounting member **6a** extends from the knuckle **6** and is disposed to face the rear surface of the caliper body module **100**. The second mounting member **6a** may be integrally formed with the knuckle **6**, or may be manufactured separately from the knuckle **6** and fixed to the knuckle **6**. The second mounting member **6a** may have a through hole formed at the end thereof and into which the second fastening member **600** is inserted. The through hole of the second mounting member **6a** may be disposed to face the second fastening member **600**, and a central axis of the through hole may be disposed on substantially the same axial line as the central axis C of the guide member **400**.

(62) The second fastening member **600** in accordance with the present embodiment may be formed in a rod shape extending horizontally from the front end of the guide pin part **410** along the longitudinal direction of the guide member **400**. A central axis of the second fastening member **600**

is disposed on substantially the same axial line as the central axis C of the guide pin part **410**. The second fastening member **600** protrudes from the rear surface of the accommodating part **140** by a predetermined distance. The protruding distance of the second fastening member **600** may be changed in design to various values according to a distance between the front end of the guide pin part **410** and the second mounting member **6a**, and the like. In a state in which the rotator **20**, the stator **30**, and the disk module **40** are assembled into the wheel **10** and then the caliper body module **100** is temporarily assembled on the disk module **40**, the second fastening member **600** is fastened to the second mounting member **6a** to fix the position of the caliper body module **100**.

(63) A second fastening reinforcing part **601** may be provided on an outer periphery of the second fastening member **600** to reinforce a fastening force of the second fastening member **600** with respect to the second mounting member **6a**. The second fastening reinforcing part **601** in accordance with the present embodiment may be formed in the form of threads that protrude from the outer circumferential surface of the second fastening member **600** and extend along the longitudinal direction of the second fastening member **600**. In such a case, threads corresponding to the second fastening reinforcing part **601** may be formed on the inner circumferential surface of the second mounting member **6a**. The shape of the second fastening reinforcing part **601** is not limited to the shape of the thread, and may be variously changed in design within the technical spirit of a shape capable of reinforcing the fastening force of the second fastening member **600** with respect to the second mounting member **6a**.

(64) In the brake apparatus **2** for a vehicle in accordance with the second embodiment of the present disclosure, the installation state of the guide pin part **410** and the second fastening member **600** may be changed according to the position of the second mounting member **6a**.

(65) FIG. **10** is a cross-sectional view illustrating a case where the second mounting part **6a** is disposed at a position different from that of FIG. **9**.

(66) Referring to FIG. **10**, when the second mounting part **6a** is located in the middle of the caliper body module **100**, more specifically, in front of the accommodating part **140**, the guide pin part **410** is formed to have a length corresponding to that of the accommodating part **140** and the front end of the guide pin part **410** is inserted into the rear surface of the accommodating part **140**.

(67) The second fastening member **600** is formed to extend from the front end of the guide pin part **410** and to protrude from the front end of the accommodating part **140** by a predetermined distance. The second fastening member **600** is inserted into the through hole of the second mounting member **6a** and is fastened to the second mounting member **6a**.

(68) The front end of the second fastening member **600** may extend toward the finger part **120** to directly penetrate the finger part **120**, or as illustrated in FIG. **10**, the second fastening member **600** may be formed separately from the guide pin part **410** and coupled to the other side of a fixing pin **7** having one side that penetrates the finger part **120**.

(69) The fixing pin **7** may movably support the pad plate part **200** while the pad plate part **200** is mounted on the outer circumferential surface thereof.

(70) Hereinafter, a configuration of a brake apparatus **3** for a vehicle in accordance with a third embodiment of the present disclosure will be described.

(71) FIG. **11** is a view schematically illustrating the brake apparatus **3** for a vehicle in accordance with the third embodiment of the present disclosure. Referring to FIG. **11**, the brake apparatus **3** for a vehicle in accordance with the third embodiment of the present disclosure includes a wheel **10**, a rotor **20**, a stator **30**, a disk module **40**, an internal braking device **50**, and an external braking device **60**.

(72) The wheel **10** is rotated in connection with the rotor **20** to be described below. For example, the wheel **10** may include a wheel connection part **11** coupled to the rotor **20** and a wheel rim part **12** formed at an end of the wheel connection part **11** and having a rim shape surrounding the stator **30**. The stator **30** may be disposed between the rotor **20** and the wheel rim part **12** due to such a shape of the wheel **10**.

(73) The stator **30** is disposed inside the wheel **10** and a tire **90** is mounted on the outer side of the wheel **10**. For example, the stator **30** may include an iron core that supports a winding and a frame that supports the iron core.

(74) The rotor **20** is disposed inside the stator **30** and is rotated when power is applied to the stator **30**. For example, the stator **30** generates a rotating magnetic field by sequentially applying power to each winding, and the rotor **20** penetrating the stator **30** is rotated along the rotating magnetic field.

(75) Meanwhile, a fixed unit **80** mounted on a vehicle body may include a fixed coupling part **81** coupled to the stator **30**, a fixed shaft part **82** formed in the center of the fixed coupling part **81** and configured to penetrate the rotor **20**, and a shaft bearing part **83** mounted on the fixed shaft part **82** and configured to rotatably support the rotor **20**.

(76) The disk module **40** is interlocked with the rotor **20**. For example, an inner circumferential surface of the disk module **40** may be coupled to the outer side of the rotor **20**, and an outer circumferential surface thereof may be disposed between the internal braking device **50** and the external braking device **60**.

(77) The internal braking device **50** is mounted on the stator **30** to provide a frictional force to one side of the disk module **40**. The external braking device **60** is mounted on the stator **30** to provide a frictional force to the other side of the disk module **40**. For example, the internal braking device **50** and the external braking device **60** may be simultaneously driven to press both surfaces of the disk module **40**, thereby providing a braking force.

(78) FIG. **12** is a view schematically illustrating the stator **30** in accordance with the third embodiment of the present disclosure. Referring to FIG. **12**, the stator **30** in accordance with the present embodiment may include a housing unit **31**, a stator body **32**, and an internal space **33**.

(79) The housing unit **31** is fixedly coupled to the fixed unit **80** and is mounted with the external braking device **60**. For example, the housing unit **31** may include a first housing part **34** disposed inside the wheel rim part **12**, a second housing part **35** extending inward from an end of the first housing part **34**, and a third housing part **36** extending from an end of the second housing part **35** and configured to surround the rotor **20**. The first housing part **34**, the second housing part **35**, and the third housing part **36** may be integrally formed, or may be separated. Not only the fixed coupling part **81** but also the external braking device **60** may be assembled to the end of the first housing part **34**. Meanwhile, a length of the first housing part **34** may be formed to be longer than that of the third housing part **36**, so that the disk module **40** may not interfere with the third housing part **36**.

(80) The stator body **32** is mounted on the housing unit **31**, is disposed to surround the rotor **20**, and rotates the rotor **20** when power is applied. For example, the stator body **32** may be embedded in the third housing part **36**.

(81) The internal space **33** is formed in the housing unit **31** and forms a space into which the internal braking device **50** is inserted. For example, the internal space **33** may be a space formed among the first housing part **34**, the second housing part **35**, and the third housing part **36**. In addition, the internal space **33** may be a groove formed in the housing unit **31**, which corresponds to the disk module **40**.

(82) FIG. **13** is a view schematically illustrating a first modification of the internal braking device **50** in accordance with the third embodiment of the present disclosure. Referring to FIG. **13**, the internal braking device **50** in accordance with the present modification includes an internal piston part **51** and an internal pad part **52**.

(83) The internal piston part **51** is mounted in the internal space **33** and has a variable length. For example, the internal piston part **51** itself may be inserted into the internal space **33** and fixedly installed in the housing unit **31**.

(84) The internal pad part **52** is mounted on the internal piston part **51** and is in surface contact with the disk module **40** according to a change in the length of the internal piston part **51**. For example, the internal pad part **52** may protrude from the internal space **33** to be in close contact with the disk

module **40** when braking is required.

(85) FIG. **14** is a view schematically illustrating a second modification of the internal braking device **50** in accordance with the third embodiment of the present disclosure. Referring to FIG. **14**, the internal braking device **50** in accordance with the present modification includes an internal case **55**, an internal variable part **56**, and an internal contact part **57**.

(86) The internal case **55** is mounted in the internal space **33** and is coupled to the external braking device **60**. For example, the internal case **55** may include a first internal case part **55a** disposed inside the first housing part **34** and configured to surround the disk module **40**, a second internal case part **55b** extending inward from an end of the first internal case part **55a** and coupled to the second housing part **35**, and a third internal case part **55c** extending laterally from the second internal case part **55b** and supported by the third housing part **36**. Meanwhile, a length of the first internal case part **55a** may be formed to be longer than that of the third internal case part **55c**, so that the disk module **40** may not interfere with the third internal case part **55c**.

(87) The internal variable part **56** is mounted in the internal case **55** and has a variable length. The internal contact part **57** is mounted on the internal variable part **56** and is in surface contact with the disk module **40** according to a change in the length of the internal variable part **56**. Since the internal variable part **56** and the internal contact part **57** correspond to the internal piston part **51** and the internal pad part **52** in FIG. **13**, detailed description thereof will be omitted.

(88) FIG. **15** is a view schematically illustrating the external braking device **60** in accordance with the third embodiment of the present disclosure. Referring to FIG. **15**, the external braking device **60** in accordance with the present modification includes an external case **61**, an external variable part **62**, and an external contact part **63**.

(89) The external case **61** is coupled to the stator **30**. For example, the external case **61** may be coupled to an end of the first housing part **34**. The external case **61** may be coupled to the first internal case part **55a**. In addition, the external case **61** may be additionally coupled to the fixed unit **80**.

(90) The external variable part **62** is mounted in the external case **61** and has a variable length. The external contact part **63** is mounted on the external variable part **62** and is in surface contact with the disk module **40** according to a change in the length of the external variable part **62**. Since the external variable part **62** and the external contact part **63** correspond to the internal variable part **56** and the internal contact part **57** in FIG. **14**, detailed description thereof will be omitted.

(91) The assembly and operation of the brake apparatus **3** for a vehicle having the above structure in accordance with the third embodiment of the present disclosure will be described as follows.

(92) The stator **30** is inserted into the wheel **10** and the internal braking device **50** is mounted on the stator **30**. Then, the rotor **20** is inserted into the stator **30**, and is disposed so that the rotor **20** is coupled to the wheel **10** and the disk module **40** connected to the rotor **20** faces the internal braking device **50**.

(93) When the above assembly is completed, the external braking device **60** is mounted on the stator **30**. At this time, the external braking device **60** is disposed to face the internal braking device **50**, and the disk module **40** is disposed between the internal braking device **50** and the external braking device **60**.

(94) When the assembly of the external braking device **60** is completed, the rotor **20** is mounted on the fixed unit **80**. At this time, the rotor **20** is rotatably supported by the shaft bearing part **83** and the stator **30** is mounted on the fixed coupling part **81**. Therefore, the stator **30** remains fixed to the vehicle body.

(95) When power is applied to the stator **30** in the above state, the rotor **20** disposed inside the stator **30** is rotated and the wheel **10** coupled to the rotor **20** is rotated, so that a vehicle is moved.

(96) When braking for the moving vehicle is required, power is applied to the internal braking device **50** and the external braking device **60**, so that the internal braking device **50** and the external braking device **60** are in close contact with the disk module **40** to limit the rotation of the rotor **20**.

(97) In the brake apparatus 3 for a vehicle in accordance with the third embodiment of the present disclosure, the internal braking device 50 and the external braking device 60 are fixed to the stator 30, so that the assembling efficiency of the vehicle body may be improved and the rigidity of the internal braking device 50 and the external braking device 60 may be secured.

(98) Although the present disclosure has been described with reference to the embodiments illustrated in the drawings, the embodiments of the disclosure are for illustrative purposes only, and those skilled in the art to which the present technology pertains will appreciate that various modifications and other equivalent embodiments are possible from the embodiments. Thus, the true technical scope of the present disclosure should be defined by the following claims.

Claims

1. A brake apparatus for a vehicle, comprising: a caliper body module provided with a finger part and a cylinder part spaced apart from each other and facing each other; a pad plate part disposed between the finger part and the cylinder part, and configured to selectively come into contact with a disk module rotated together with a rotor to generate a braking force; a piston module installed in the cylinder part to be movable forward and backward and configured to press the pad plate part; a guide member disposed on both sides of the caliper body module and configured to movably support the caliper body module; and a first fastening member extending from the guide member and fastened to a first mounting member provided in a stator.
2. The brake apparatus according to claim 1, wherein the first fastening member is disposed on substantially the same axial line as the guide member.
3. The brake apparatus according to claim 2, wherein the first fastening member protrudes to an outside of the finger part along a longitudinal direction of the guide member.
4. The brake apparatus according to claim 2, further comprising a first fastening reinforcing part provided on an outer periphery of the first fastening member to reinforce a fastening force of the first fastening member with respect to the first mounting member.
5. The brake apparatus according to claim 1, wherein a central axis of the guide member is disposed at substantially the same height as a center of gravity of the caliper body module.
6. The brake apparatus according to claim 5, wherein the guide member comprises: a guide pin part configured to penetrate the caliper body module and disposed in a direction parallel to a forward and backward movement direction of the piston module; a friction reduction part configured to surround the guide pin part; a movement part slidably connected to the friction reduction part and configured to move the caliper body module by a pressing force of the piston module; and a blocking part configured to block foreign substances from entering between the caliper body module and the friction reduction part.
7. The brake apparatus according to claim 6, wherein the guide pin part has one side connected through the finger part.
8. The brake apparatus according to claim 6, wherein the guide pin part movably supports the pad plate part.
9. A brake apparatus for a vehicle, comprising: a caliper body module provided with a finger part and a cylinder part spaced apart from each other and facing each other; a pad plate part disposed between the finger part and the cylinder part, and configured to selectively come into contact with a disk module rotated together with a rotor to generate a braking force; a piston module installed in the cylinder part to be movable forward and backward and pressing the pad plate part; a guide member disposed on both sides of the caliper body module and configured to movably support the caliper body module; and a fastening member extending from the guide member and fastened to a mounting member provided on a knuckle.
10. The brake apparatus according to claim 9, wherein the fastening member is disposed on substantially the same axial line as the guide member.

11. The brake apparatus according to claim 10, wherein the fastening member protrudes to an outside of the caliper body module along a longitudinal direction of the guide member.
 12. The brake apparatus according to claim 10, further comprising a fastening reinforcing part provided on an outer periphery of the fastening member to reinforce a force of the fastening member with respect to the mounting member.
 13. The brake apparatus according to claim 9, wherein a central axis of the guide member is disposed at substantially the same height as a gravity axis of the caliper body module.
 14. The brake apparatus according to claim 13, wherein the guide member comprises: a guide pin part configured to penetrate the caliper body module and disposed in a direction parallel to a forward and backward movement direction of the piston module; a friction reduction part configured to surround the guide pin part; a movement part slidably connected to the friction reduction part and configured to move the caliper body module by a pressing force of the piston module; and a blocking part configured to block foreign substances from entering between the caliper body module and the friction reduction part.
 15. The brake apparatus according to claim 14, wherein the guide pin part has one side connected through the finger part.
 16. A brake apparatus for a vehicle, comprising: a stator; a rotor disposed inside the stator and rotated when power is applied to the stator; a wheel rotated in connection with the rotor; a disk module interlocked with the rotor; an internal braking device mounted on the stator and configured to provide a frictional force to one side of the disk module; and an external braking device mounted on the stator and configured to provide a frictional force to the other side of the disk module, wherein the stator comprises: a housing unit mounted with the external braking device and fixedly coupled to a fixed unit; a stator body mounted on the housing unit and configured to surround the rotor and rotate the rotor when power is applied; and an internal space in the housing unit configured to accommodate the internal braking device.
 17. The brake apparatus according to claim 16, wherein the internal braking device comprises: an internal piston part mounted in the internal space and having a variable length; and an internal pad part mounted on the internal piston part and being in surface contact with the disk module according to a change in the length of the internal piston part.
 18. The brake apparatus according to claim 16, wherein the internal braking device comprises: an internal case mounted in the internal space and coupled to the external braking device; an internal variable part mounted in the internal case and having a variable length; and an internal contact part mounted on the internal variable part and being in surface contact with the disk module according to a change in the length of the internal variable part.
 19. The brake apparatus according to claim 16, wherein the external braking device comprises: an external case coupled to the stator; an external variable part mounted in the external case and having a variable length; and an external contact part mounted on the external variable part and being in surface contact with the disk module according to a change in the length of the external variable part.
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